

Class-D Homework

- For each lecture there are three homework assignment options that are comparable in complexity. For each lecture, choose one and prepare slide set for a 10-15min presentation. So:
 - for lecture 1 choose either 1A, 1B or 1C,
 - for lecture 2 choose either 2A, 2B or 2C,
 - for lecture 3 choose either 3A, 3B or 3C
- When showing graphs, make sure the axes are properly labeled with readable text. Preferably, use white background and make sure the traces are labeled and clearly visible.
- Show the steps you make to arrive at your results and check if your results match your expectation.

Homework Assignment 1A

Amplifier Classes & Pulse Width Modulation

1. Calculate the inductor and capacitor values of a lowpass LC filter with the following specification: load impedance 4Ω , attenuation @ 1MHz >50dB, quality factor $0.7 < Q < 1.0$. The inductor and capacitor values have to be selected from an E12-series and a tolerance of $\pm 10\%$ for the inductor and $\pm 20\%$ for the capacitor has to be taken into account.
Create a schematic in Cadence Virtuoso and verify your results.
 - what are the realized nominal corner frequency and quality factor ?
 - make a bode-plot of the frequency transfer to verify
2. In Cadence Virtuoso create a PWM generator with elements from **analogLib** and **ahdlLib** that makes a double-sided PWM signal with a carrier frequency $f_{\text{PWM}} = 1\text{MHz}$. Let the output V_{PWM} switch between -10V and +10V. Connect the LC filter and 4Ω load.
 - run simulations using sinewave inputs with 0.1 [FS] amplitude and frequencies 1kHz, $f_{\text{PWM}}+1\text{kHz}$, $2f_{\text{PWM}}+1\text{kHz}$
 - inspect the signal after the LC-filter: explain your observations
 - make a frequency spectrum plot of the signal after the LC-filter using the Spectrum function in the **ViVa** waveform viewer and explain your observations
 - increase the amplitude to 0.9 [FS]: describe and explain your observations
3. In Cadence Virtuoso add a second PWM generator and inductor to create two-phase PWM
 - run simulations using a DC input signal for both the single-phase and two-phase PWM configuration
 - make a plot of the peak-to-peak ripple current for both systems versus modulation index. describe and explain your observations
 - verify the peak values with calculations

Homework Assignment 1B

Amplifier Classes & Pulse Width Modulation

1. Calculate the inductor and capacitor values of a lowpass LC filter with the following specification: load impedance 8Ω , attenuation @ 600kHz >50dB, passband (20kHz) ripple between ± 1 dB. The inductor and capacitor values have to be selected from an E12-series and a tolerance of $\pm 10\%$ for the inductor and $\pm 20\%$ for the capacitor has to be taken into account. Create a schematic in Cadence Virtuoso and verify your results.
 - what are the realized nominal corner frequency and quality factor ?
 - make a bode-plot of the frequency transfer to verify
2. In Cadence Virtuoso create a PWM generator with elements from **analogLib** and **ahdlLib** that makes a **single-sided** PWM signal with a carrier frequency $f_{\text{PWM}} = 600\text{kHz}$. Let the output V_{PWM} switch between -12V and +12V. Connect the LC filter and 8Ω load.
 - run a simulation with a 1kHz sinewave input and 0.1 [FS] amplitude.
 - make a frequency spectrum plot of the signal after the LC-filter using the Spectrum function in the **VivA** waveform viewer
3. Modify the PWM generator to make a **double-sided** PWM signal.
 - run a simulation with a 1kHz sinewave input and 0.1 [FS] amplitude.
 - make a frequency spectrum plot of the signal after the LC-filter using the Spectrum function in the **VivA** waveform viewer
 - compare the frequency spectra of the single-sided and double-sided PWM signals. describe and explain your observations
4. In Cadence Virtuoso add a second PWM generator and LC filter to create a Bridge-Tied-Load (BTL) configuration
 - run simulations with a 1kHz sinewave inputs signal and 0.7 [FS] amplitude in both **BD** and **AD** modulation
 - make a frequency spectrum plot of both the **differential-mode** and **common-mode** signals after the LC-filter
 - compare the spectra of BD and AD. describe and explain your observations

Homework Assignment 1C

Amplifier Classes & Pulse Width Modulation

1. Calculate the inductor and capacitor values for a lowpass LC filter with a corner frequency of 30kHz and a quality factor of 1 when a 2Ω load impedance is attached. Select the nearest values from an E12-series, create a schematic in Cadence Virtuoso and verify your results
 - what are the realized corner frequency and quality factor ?
 - make a bode-plot of the frequency transfer to verify
2. In Cadence Virtuoso create a PWM generator with element from **analogLib** and **ahdlLib** that makes a double-sided PWM signal with a carrier frequency $f_{\text{PWM}} = 256\text{kHz}$. Let the output V_{PWM} switch between -5V and +5V. Connect the LC filter and 2Ω load.
 - run a simulation with a 1kHz sinewave input and 0.1 [FS] amplitude.
 - make a frequency spectrum plot of the signal after the LC-filter using the Spectrum function in the **VivA** waveform viewer
 - rerun the simulation with increasing accuracy by changing the **reltol** parameter (ADE->Simulation->Options), e.g. reltol = 1e-3, 1e-4, 1e-5 etc...
 - compare the frequency spectra of these simulations:
 - determine the SINAD (Signal-to-Noise and Distortion) in the **audio bandwidth** for the different reltol settings. (tip: use **spectrumMeasurement** expression)
 - make a plot of **SINAD** versus **reltol**: describe and explain your observations.
3. Insert an ideal Sample&Hold (**sah_ideal** from **ahdlLib**) between the input signal and the PWM generator. Use sample frequency equal to f_{PWM} .
 - run a series of simulations with a sinewave input with frequencies {1, 2, 4, 8} [kHz] and amplitudes {0.1, 0.2, 0.4, 0.8} [FS]
 - examine and analyze the frequency spectra of these simulations: describe and explain your observations
 - make plots of **SINAD** versus **signal frequency** signal and **SINAD** versus **signal amplitude**: describe and explain your observations
4. Move the Sample & Hold between the triangle reference and the PWM generator. Increase the sample frequency to $64 \times f_{\text{PWM}}$
 - run a series of simulations with a sinewave input with frequencies 1kHz and amplitudes {0.1, 0.2, 0.4, 0.8} [FS]
 - examine the time domain waveforms of the signal after the LC-filter: describe and explain your observations
 - examine and analyze the frequency spectra of these simulations: describe and explain your observations
 - make plot of **SINAD** versus **signal amplitude**: describe and explain your observations
 - repeat the simulations with a sample frequency of $1024 \times f_{\text{PWM}}$: describe and explain your observations